

SARFARAZ JELIL

Bengaluru, Karnataka, India

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EDUCATION

Ph.D. in Speech Processing Indian Institute of Technology Guwahati, India	February 2025
M.Tech. in Information Technology North-Eastern Hill University, Shillong, India	June 2015
B.Tech. in Information Technology North-Eastern Hill University, Shillong, India	June 2011

RESEARCH & ACADEMIC EXPERIENCE

Indian Institute of Technology Guwahati <i>Doctoral Researcher</i>	Guwahati, India <i>Jan 2016 – Jan 2025</i>
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- Conducted doctoral research on replay-attack detection for automatic speaker verification, developing a source-filter decomposition framework to separately analyze source and filter information in speech signals and exploit their complementary characteristics for robust spoofing detection.
- Developed novel source- and filter-based acoustic features and investigated their individual and combined representations for improving replay-attack detection performance.
- Designed and evaluated systems using classical statistical classifiers and deep-learning approaches, providing comparative analysis of different modeling techniques for speech anti-spoofing.
- Evaluated approaches on established speaker-recognition and spoofing datasets and participated in speaker-recognition challenges, including the ASVspoof 2017 challenge, in collaboration with other researchers.
- Published peer-reviewed research in speech processing and speaker verification, including publications in *IEEE Signal Processing Letters* and presentations at *Interspeech* 2018 and 2019.
- Mentored one M.Tech. and one B.Tech. student during doctoral research, providing technical guidance on research experimentation and speech/ML problems.

North-Eastern Hill University <i>Laboratory Engineer — Government of India-funded research project</i>	Shillong, India <i>Sep 2012 – Jun 2015</i>
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- Worked on a Government of India-funded research project under the Ministry of Communications and Information Technology, “Development of Speech-Based Multilevel Person Authentication System,” focused on practical speech-biometric authentication.
- Helped establish the speech technology laboratory and experimental infrastructure, including a voice server and equipment for speech-data collection and speaker-verification experiments.
- Developed an IVR-based system for collecting speech data over telephone networks, contributing to the creation of a speech database involving approximately 300 users.
- Developed a real-time voice-based attendance and authentication system in C, performing speech feature extraction and speaker classification for online speaker authentication.
- Conducted research on voice-password-based speaker verification using MFCC features and DTW, and contributed to integrating voice-password, text-dependent, and text-independent speaker-verification components into the multilevel authentication framework.

TEACHING EXPERIENCE

Indian Institute of Technology Guwahati
Teaching Assistant / Academic Teaching Support

Guwahati, India
Jan 2016 – Jun 2019

- Conducted tutorial sessions for undergraduate students in *Signals and Systems*, reinforcing faculty-led lectures through worked examples, problem-solving exercises, conceptual discussions, and clarification of student questions.
- Supported laboratory instruction for *Signals and Systems*, *Basic Electronics*, and *Digital Circuits*, guiding students through experiments, measurement procedures, troubleshooting, and interpretation of results.
- Assisted with assignments, laboratory work, grading, assessment, and course records, providing feedback to students on their academic work.
- Provided individual and small-group guidance to students in problem-solving and the practical application of concepts in signals, electronics, and digital systems.

RESEARCH INTERESTS

AI for Organizational Decision-Making; Reliable and Evidence-Based AI; Generative and Agentic AI; Human–AI Decision Systems; AI for Business and Financial Services; Applied Machine Learning.

INDUSTRY AI/ML EXPERIENCE

Wissen Technology Pvt. Ltd.
Senior Software Engineer (AI)

Bengaluru, India
Feb 2026 – Present

- Developing and supporting an enterprise agentic-AI system in active production use for a global financial-services client, enabling service professionals and financial advisors to retrieve information from complex workflow-management systems through natural-language queries.
- Designing an agentic workflow that classifies user intent, maps natural-language parameters to enterprise data fields, generates Elasticsearch queries, and invokes backend APIs and tools to retrieve workflow information across KYC, payments, ACH, wire transfers, and document-processing workflows.
- Designing structured data flows between the agent, retrieval services, and frontend, ensuring deterministic delivery of retrieved information while using the LLM primarily for natural-language response generation.
- Developing evaluation and optimization approaches for field-level and query-level accuracy, while improving latency and reliability by reducing unnecessary LLM processing of large tool responses.
- Working across Python, OpenAI Agents SDK, LLMs, Elasticsearch, Pydantic structured outputs, and API-based tool integration.
- Contributing to AI/ML talent evaluation by conducting technical interviews and assessing candidates across machine learning, Generative AI, Python, and AI engineering competencies.

British Telecom (BT Group)

Machine Learning Engineer / Design Engineer Professional (Machine Learning) Feb 2023 – Feb 2026

Bengaluru, India

- Developed an enterprise retrieval-augmented generation (RAG) application in Python for conversational access to technical design documents and specifications, using locally hosted open-source LLMs including DeepSeek and Mistral through Ollama.
- Worked across the RAG/LLM pipeline, including document processing, knowledge-base construction, retrieval, prompt design, model selection, response generation, and evaluation, with emphasis on response quality and relevance.

- Developed an LLM-based sentiment-analysis and insight-generation workflow as a proof of concept for company reviews, identifying strengths, weaknesses, and actionable themes.
- Designed and developed a machine learning and deep-learning pipeline for predicting task dependencies in broadband service delivery, working across feature engineering, model development, evaluation, and application integration.
- Exposed the task-link prediction model through a Java Spring Boot API and integrated it with an existing enterprise application, supporting production deployment.
- Worked across the AI application lifecycle, spanning machine learning, Generative AI, retrieval, evaluation, API integration, and deployment.

BlueOptima

Associate Machine Learning Data Engineer

Bengaluru, India

Sep 2022 – Jan 2023

- Developed a Python-based system to identify and track open-source code within enterprise codebases, supporting software-license compliance and mitigation of associated legal risks.
- Applied graph-based machine learning techniques in Enterprise Structure Automation to identify relationships among code repositories and improve understanding of enterprise codebase structure.

Bay Area Advanced Analytics

Data Scientist

Guwahati, India

Jul 2019 – Sep 2022

- Developed deep-learning computer-vision systems for surgical video and medical imaging, focusing on automated detection and segmentation of anatomical structures and pathologies.
- Experimented with multiple CNN architectures for shoulder and knee anatomy segmentation and evaluated their suitability for surgical applications.
- Worked directly with surgeons and clinical experts to understand relevant anatomy and establish labeling criteria, and contributed to initial annotation of surgical datasets.
- Developed anatomy and pathology detection models for surgical video and contributed to software capabilities for automated surgical evidence and report generation.
- Developed YOLO and Faster R-CNN models for surgical probe detection in medical images and contributed to their integration into surgical software.
- Developed algorithms for curvilinear measurement of anatomical structures in surgical images.
- Developed the anatomy-detection component underlying the patented Surgery Evidence Report Generation system (WO2023220674A2).

PUBLICATIONS

Journal Publications

1. S. Jelil, R. Sinha, and S. R. M. Prasanna, "Spectro-Temporally Compressed Source Features for Replay Attack Detection," *IEEE Signal Processing Letters*, vol. 31, pp. 721–725, 2024.
2. R. K. Das, S. Jelil, and S. R. M. Prasanna, "Exploring Text-Constraint Models and Source Information for Long-Enrollment with Short-Test Speaker Verification," *Circuits, Systems, and Signal Processing*, vol. 38, pp. 1775–1792, 2018.
3. R. K. Das, S. Jelil, and S. R. M. Prasanna, "Multi-Style Speaker Recognition Database in Practical Conditions," *International Journal of Speech Technology*, vol. 21, no. 3, pp. 409–419, 2018.
4. A. Hore, S. R. Nirmala, R. K. Das, S. Jelil, and S. R. M. Prasanna, "Significance of Source Information for Text Dependent Speaker Verification," *International Journal of Image, Graphics and Signal Processing*, vol. 10, pp. 42–49, 2018.

5. R. K. Das, S. Jelil, and S. R. M. Prasanna, “Development of Multi-Level Speech based Person Authentication System,” *Journal of Signal Processing Systems*, pp. 1–13, 2016.

Conference Publications

1. S. Jelil, A. Shrivastava, R. K. Das, S. R. M. Prasanna, and R. Sinha, “SpeechMarker: A Voice based Multi-Level Attendance Application,” in *Interspeech*, 2019.
2. S. Jelil, S. Kalita, S. R. M. Prasanna, and R. Sinha, “Exploration of Compressed ILPR Features for Replay Attack Detection,” in *Interspeech*, 2018.
3. S. Pandey, S. Jelil, S. R. M. Prasanna, and H. S. Shekhawat, “Speaker Identification using Tensor Decomposition of Acoustic Models,” in *TENCON*, 2018, pp. 1484–1488.
4. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “Spoof Detection Using Source, Instantaneous Frequency and Cepstral Features,” in *Interspeech*, 2017.
5. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “IITG-Indigo System for NIST 2016 SRE Challenge,” in *Interspeech*, 2017.
6. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “Role of Voice Activity Detection Methods for Speakers in Wild Challenge,” in *National Conference on Communications (NCC)*, 2017.
7. S. Jelil, R. K. Das, and S. R. M. Prasanna, “Significance of Constraining Text in Limited Data Text-Independent Speaker Verification,” in *SPCOM*, 2016.
8. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “AMRITATCS-IITGUWAHATI Combined System for SITW Challenge,” in *TENCON*, 2016.
9. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “Exploring Session Variability and Template Aging in Speaker Verification for Fixed Phrase Short Utterances,” in *Interspeech*, 2016.
10. S. Jelil, R. K. Das, S. R. M. Prasanna, and R. Sinha, “Speaker Verification Using Gaussian Posteriorgrams on Fixed Phrase Short Utterances,” in *Interspeech*, 2015.
11. S. Jelil, R. K. Das, and S. R. M. Prasanna, “Exploring Speaker Modeling Techniques for Short Pass-Phrase based Person Authentication System,” in *I3CS*, 2015.
12. S. Jelil, R. K. Das, and S. R. M. Prasanna, “Line, Word, and Character Segmentation of Manipuri Machine Printed Text,” in *CICN*, 2014.
13. S. Jelil, R. K. Das, and S. R. M. Prasanna, “Comparative Evaluation of Feature Normalization Techniques for Voice Password based Speaker Verification,” in *NCVPRIPG*, 2013.

PATENT

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- S. Jelil et al., “**Surgery Evidence Report Generation**,” **WO2023220674A2**, February 2024.
 - Contributed to the development of the anatomy-detection component of the patented system, which uses neural networks to analyze surgical video for identification of anatomical structures, pathologies, and surgical activities.

AWARDS & FELLOWSHIPS

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- **Visvesvaraya Ph.D. Fellowship**, Ministry of Electronics and Information Technology, Government of India, 2016–2019.
 - **Microsoft Travel Award**, 2017 – Awarded to attend and present research at Interspeech 2017, Stockholm, Sweden.

- **ISCA Travel Award**, 2015 – Awarded to attend and present research at Interspeech 2015, Dresden, Germany.

PRESENTATIONS

- **Interspeech 2018** – Hyderabad, India, “Exploration of Compressed ILPR Features for Replay Attack Detection”
Poster Presentation
- **Interspeech 2017** – Stockholm, Sweden, “Spoof Detection Using Source, Instantaneous Frequency and Cepstral Features”
Oral Presentation
- **National Conference on Communications (NCC) 2017** – Chennai, India, “Role of Voice Activity Detection Methods for the Speakers in the Wild Challenge”
Oral Presentation
- **Interspeech 2015** – Dresden, Germany, “Speaker Verification Using Gaussian Posteriorgrams on Fixed Phrase Short Utterances”
Poster Presentation
- **National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG) 2013** – Jodhpur, India, “Comparative Evaluation of Feature Normalization Techniques for Voice Password based Speaker Verification”
Poster Presentation

TECHNICAL SKILLS

- **Programming:** Python, MATLAB.
- **AI/ML Frameworks:** PyTorch, TensorFlow, OpenAI Agents SDK, LangChain.
- **AI Engineering & MLOps:** FastAPI, Elasticsearch, Hugging Face, Docker, Git, MLflow, TeamCity, Grafana.

REFERENCES

- **Prof. S. R. Mahadeva Prasanna**
Director, Indian Institute of Information Technology Dharwad,
Karnataka, India – 580009
Email: prasanna@iiitdwd.ac.in
- **Prof. Rohit Sinha**
Professor, Department of Electronics and Electrical Engineering,
Indian Institute of Technology Guwahati, Assam, India – 781039
Email: rsinha@iitg.ac.in
- **Dr. Rohan Kumar Das**
Research and Development Manager, Fortemedia,
Singapore 189352
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